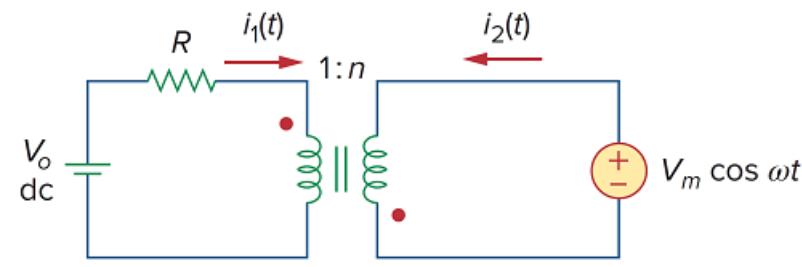


Problem

In the ideal transformer circuit of Fig. 13.109, find $i_1(t)$ and $i_2(t)$.

Figure 13.109



Step-by-step solution

Step 1 of 6

Refer to figure 13.109 for the circuit in the textbook.

Consider that a DC source is acting alone as shown in figure 1.

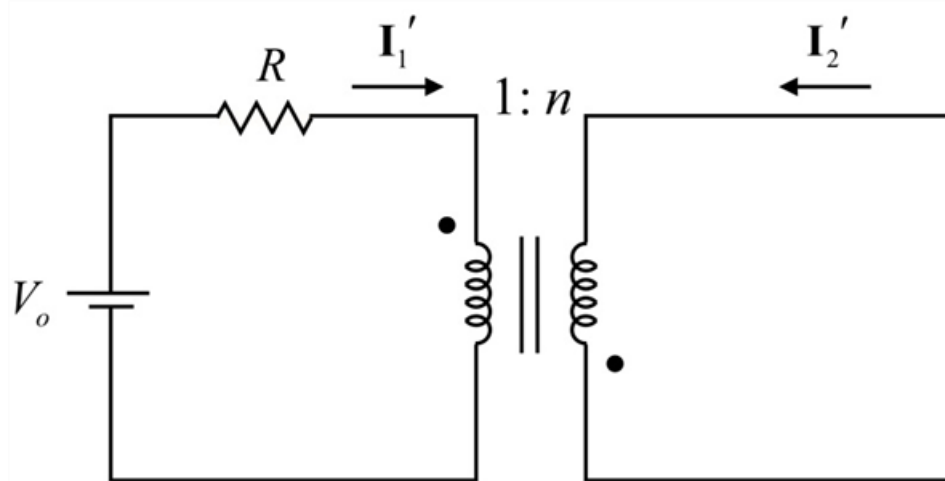


Figure 1

Step 2 of 6

Consider that there's no secondary impedance reflecting to primary side. Therefore,

$$Z_R = 0 \, \Omega$$

Determine the expression for primary current I_1' for DC source.

$$\begin{aligned} I_1' &= \frac{V_o}{Z_1 + Z_R} \\ &= \frac{V_o}{R + 0} \\ &= \frac{V_o}{R} \end{aligned}$$

Therefore,

$$i_1'(t) = \frac{V_o}{R}$$

Determine the expression for secondary current I_2' for DC source.

$$\begin{aligned} I_2' &= \frac{1}{n} I_1' \\ &= \frac{1}{n} \left(\frac{V_o}{R} \right) \end{aligned}$$

Therefore,

$$i_2'(t) = \frac{V_o}{nR}$$

Step 3 of 6

Consider that a single AC source is acting alone as shown in figure 2.

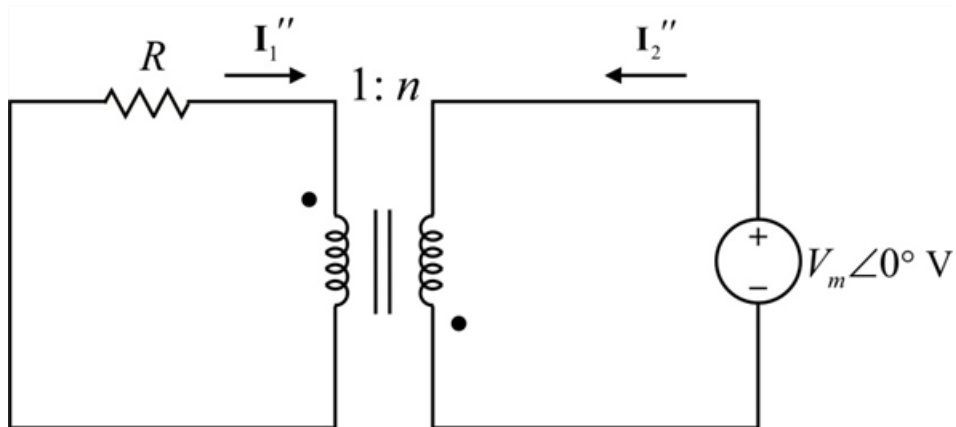


Figure 2

Step 4 of 6

Here, the primary impedance reflecting to the secondary side. Therefore,

$$\begin{aligned} Z_R &= n^2 Z_s \\ &= n^2 R \end{aligned}$$

Determine the expression for primary current $\mathbf{I}_2''$ for AC source.

$$\begin{aligned} \mathbf{I}_2'' &= \frac{V_m \angle 0^\circ}{Z_1 + Z_R} \\ &= \frac{V_m \angle 0^\circ}{0 + n^2 R} \\ &= \frac{V_m \angle 0^\circ}{n^2 R} \end{aligned}$$

Therefore,

$$i_2''(t) = \frac{1}{n^2 R} (V_m \cos \omega t)$$

Step 5 of 6

Determine the expression for primary current $\mathbf{I}_1''$ for AC source.

$$\begin{aligned} \mathbf{I}_1'' &= n \mathbf{I}_2'' \\ &= n \left(\frac{V_m \angle 0^\circ}{n^2 R} \right) \\ &= \frac{V_m \angle 0^\circ}{n R} \end{aligned}$$

Therefore,

$$i_1''(t) = \frac{1}{n R} (V_m \cos \omega t)$$

Step 6 of 6

Apply the superposition theorem.

$$\begin{aligned} i_1(t) &= i_1'(t) + i_1''(t) \\ &= \frac{V_o}{R} + \frac{1}{n R} (V_m \cos \omega t) \\ &= \frac{1}{R} \left[V_o + \frac{1}{n} (V_m \cos \omega t) \right] \text{ A} \end{aligned}$$

Also,

$$\begin{aligned} i_2(t) &= i_2'(t) + i_2''(t) \\ &= \frac{V_o}{n R} + \frac{1}{n^2 R} (V_m \cos \omega t) \\ &= \frac{1}{n R} \left[V_o + \frac{1}{n} (V_m \cos \omega t) \right] \text{ A} \end{aligned}$$

Therefore, the expressions for currents are,

$$\boxed{\begin{aligned} i_1(t) &= \frac{1}{R} \left[V_o + \frac{1}{n} (V_m \cos \omega t) \right] \text{ A} \\ i_2(t) &= \frac{1}{n R} \left[V_o + \frac{1}{n} (V_m \cos \omega t) \right] \text{ A} \end{aligned}}$$